INCH-POUND

MIL-S-44384
16 November 1989

MILITARY SPECIFICATION

SUIT, CONTAMINATION AVOIDANCE, LIQUID PROTECTION (SCALP)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid contamination avoidance suit made from high density polyethylene fiber and polyethylene coatings, and consisting of a hooded poncho, trousers, and footwear covers.

1.2 Classification. The suit shall be of the following sizes as specified (see 6.2).

Poncho/trousers	Footwear covers
Small	Small
Medium/Large	Medium
X-large/XX-large	Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

L-P-512	-	Plastic Sheet (Sheeting), Polyethelene
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-26		Bag, Polyethylene
PPP-B-636		Boxes, Shipping, Fiberboard

MILITARY

MIL-F-20652/1	Eyelets, Metallic Rolled Flange Type; and Eyelet Washer
MIL-L-35078	Loads, Unit, Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for
MIL-C-43307	Cord, Nylon, Solid Braid, General Purpose
MIL-T-43548	Thread, Polyester Core: Cotton, Rayon, or Polyester covered
MIL-C-43701	Cord, Elastic, Nylon

STANDARDS

FEDERAL

FED-STD-101	-	Test Procedures for Packaging Materials
FED-STD-191	-	Textile Test Methods
FED-STD-595		Colors
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147		Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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CHEMICAL RESEARCH AND DEVELOPMENT CENTER

CRDC-SP-84010 - Laboratory Method for Evaluating Protective Clothing Systems Against Chemical Agents

(Copies are available from the Chemical Research and Development Center, ATTN: SMCCR-PPI, Aberdeen Proving Ground, MD 21010.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Technical Manual of the American Association of Textile Chemists and Colorists

Method 127 - Hydrostatic head, water resistance testing

Method 76 - Electrical resistivity to testing

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1117 - Standard Methods of Testing Non-Woven Fabrics
D 1682 - Breaking Load and Elongation of Textile Fabrics
D 2103 - Standard Specification for Polethylene Film and Sheeting
D 2261 - Tearing Strength of Woven Fabrics by The Tongue Method
D 3776 - Mass Per Unit Area (Weight) of Woven Fabric
D 3786 - Hydraulic Bursting Strength of Knitted Goods and Nonwoven Fabric

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Base cloth. The base cloth shall be a nonwoven, spunbonded fabric made from 100 percent high density polyethylene fibers. The fabric shall be formed by spinning very fine polyethylene fibers and then bonding them together with heat and pressure. No binders, sizers, or fillers shall be used. The base cloth shall conform to the requirements specified in table I.

3.3.2 Coating compound. The coating compound shall be a low density polyethylene, suitably compounded and pigmented such that the finished coating will be camouflage green 483.

3.3.3 Coated cloth. The coated cloth shall be the base cloth coated on one side with the coating compound. The base cloth shall conform to the requirements specified in table I.

TABLE I. Nonwoven base cloth and coated cloth requirements

Component	Characteristic	Requirement	Test method
Base cloth	Weight, oz/sq.yd	1.10 - 1.20	ASTM D 3776 1/
	Breaking load, cut strip method		
	Machine dir.(lbs/in)	5.2 - 9.8	ASTM D 1117 2/
	Transverse dir.(lbs/in)	6.7 - 11.3	ASTM D 1117 2/
	Tongue tear strength		
	Machine dir.(lbs/in)	1.4 - 3.4	ASTM D 2261 3/
	Transverse dir.(lbs/in)	1.4 - 3.4	ASTM D 2261 3/
	Water resistance hydrostatic head, in.	40	AATCC 127
	Electrical resistivity Log R	7.0 - 8.6	AATCC 76 4/

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TABLE I. Nonwoven base cloth and coated cloth requirements (cont'd)

Component	Characteristic	Requirement	Test method
Coated cloth	Thickness, inch	0.006 - 0.007	ASTM D 2103
	Weight, oz/sq yd	2.0 - 2.4	ASTM D 3776
	Min tensile strength (lb/in) 5/		
	Initial:		
	Machine dir.	9.5	ASTM D 1682
	Transverse dir.	10.0	ASTM D 1682
	After oven aging:		
	Machine dir.	9.0	FED-STD-191 - 5850 9/
	Transverse dir	9.5	FED-STD-191 - 5850 9/
	Tear strength, lb		
	Machine and transverse dir.	3.0	ASTM D 2261
	Bursting strength, psi	19.0	ASTM D 3786
	Resistance to leakage:		
	Initial	Pass/Fail	FED-STD-191 - 5516 6/
	After cold crack	Pass/Fail	FED-STD-191 - 5874 7/ and 5516 6/
	Resistance to blocking rating	No. 2	FED-STD-191 - 5872
	Resistance to toxicological agent, minutes 8/:		
	Mustard (HD)	60	CRDC-SP-84010
	Thickened (GD)	60	CRDC-SP-84010
	Nerve (VX)	60	CRDC-SP-84010

1/ Square yard.

2/ Cut strip; CRE tensile test, crosshead speed 2 in/min, gauge length 5 inches.

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3/ Run at crosshead speed of 5 in/min and 1 inch gauge length.

4/ Converted to Log E.

5/ One-inch wide cut strip.

6/ The polyethylene coated side shall contact the water. The hydrostatic pressure shall be 50 cm for 5 minutes.

7/ Conditioned for 30 minutes at -60°F prior to testing.

8/ Chemical Research Development and Engineering Center Test.

9/ Temperature at 180°F for 7 days.

3.3.3.1 Strapping. The strapping for sealing all seams and stitching shall be strips of the coated cloth specified in 3.3.3 applied as specified in 3.6.2.

3.3.3.2 Adhesive tape. The adhesive tape for sealing the folded edges of strapping and then sealing the strapping to the seams or stitching shall be any type of thermoplastic adhesive, minimum 4 mils. thick, conforming to and from test requirements in table VI.

3.3.4 Sole material. The material for footwear soles shall be an embossed, low density polyethylene, type I, class H, grade 2 conforming to L-P-512, with a thickness of 0.012 (± 0.002) inch.

3.3.5 Cord, elastic. The draw cord for the trouser waist shall be Olive Drab 7, elastic nylon, type II, conforming to MIL-C-43701.

3.3.6 Cord, nylon. The draw cord for the poncho hood shall be solid braid nylon, Olive Drab 7, 3/32 inch diameter, conforming to MIL-C-43307.

3.3.7 Latex closures. The closures at the wrists, trouser ankles, and footwear ankles shall be natural latex, 3/8 inch wide, 0.45 inch thick and natural in color.

3.3.8 Cord locks. The cord locks for the drawcords shall be black, ITW Nexus Mini Cord-Lok, or equivalent (see 6.6).

3.3.9 Eyelets. The eyelets and washers for hood and waist drawcord shall be BBE114 and BBW101, respectively, conforming to MIL-E-20652/1.

3.3.10 Thread. The thread for stitching and seaming shall be polyester, white, size 50, 2 ply, conforming to MIL-T-43548.

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3.3.11 Labels. Each component of the suit shall have a combination size and identification label conforming to class 14 of DDD-L-20 and shall contain the following information:

Nomenclature
Size
Name of contractor
Date of manufacture
Number of contract

3.3.12 Instruction legend. The donning and doffing instruction legend shall be as specified below. The marking shall be applied to the unit pack polyethylene bag in accordance with 5.4.1.

DONNING PROCEDURES FOR THE SCALP

Step 1 Unpack SCALP footwear covers, trousers, and poncho. Inspect each piece for holes and tears. If any damage to the SCALP is present, exchange it for a new one.

Step 2 Place SCALP footwear covers over boots and pull sides up over ankles.

Step 3 Step into trousers and pull them up as far as possible. Make sure pantlegs go over the SCALP footwear covers. Hold drawcord and pull black Cordloc towards you to tighten.

Step 4 Don mask.

Step 5 Place arms into sleeves of SCALP poncho, and pull down over head and top of trousers.

Step 6 Fit hood around mask eyepieces and filter inlets. Pull drawcord and tighten both Cordlocs by pulling toward the mask until tight. Make sure hood is still fitted around eyepieces and filter inlet.

Step 7 Put on gloves and tuck ends underneath sleeves of poncho.

DOFFING PROCEDURES FOR THE SCALP

Step 1 Establish a hot line and face away from it, e.g., your back will face the hot line.

Step 2 Use detactor paper to determine areas of gross liquid contamination on your SCALP. CAUTION: TAKE SPECIAL CARE TO AVOID TOUCHING THESE AREAS DURING SCALP DOFFING.

Step 3 Check exposed protective mask with aid of a buddy, if possible.

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Step 4 Decontaminate gloves, lower portion of sleeves, and around ankles.

Step 5 Recheck and decontaminate protective mask and front edge of SCALP hood.

Step 6 Place one hand on the voice mitter to hold the mask tight to the face. Loosen Cordlocs and drawcords on SCALP around protective mask, and loosen hood of SCALP away from protective mask. Hold sleeve of SCALP around wrist area with one hand and pull arm inside SCALP poncho. Using free hand and arm inside the SCALP, reach inside other sleeve and grasp sleeve wrist area and pull other arm through to the inside.

Step 7 Bend body forward at the waist, and using free hands inside of SCALP, push SCALP up and off away from user's body into hot area.

Step 8 Turning 180 degrees, face the hot line. Loosen cordloc and drawcord of SCALP trousers. Keeping hands inside SCALP, push SCALP down to ankle area. Using hand, remove one foot from SCALP trousers and footwear covers and place on "clean side" of hot line. Using the same procedure, remove the SCALP from the remaining foot. Proceed to the clean area.

3.3.13 Fit and size prediction legend. The fit and size prediction legend shall be as specified below and shall be applied to the unit pack polyethylene bag in accordance with 5.4.1.

SUIT, CONTAMINATION AVOIDANCE, AND LIQUID PROTECTIVE

SIZE PREDICTION CHART
(Poncho & Trousers)

Waist size	Over underwear		Over coat and trousers		Over coat and trousers/field wear		Over parka and trousers/arctic wear	
	S,CP	SCALP	S,CP	SCALP	S,CP	SCALP	S,CP	SCALP
Up to 19.0	XXX-S	S	XXX-S	S	XX-S	S	X-S	S
19.1 - 23.0	XXX-S	S	XX-S	S	X-S	S	S	S
23.1 - 27.0	XX-S	S	X-S	S	S	S	M	M/L
27.1 - 31.0	X-S	S	S	S	M	M/L	L	M/L
31.1 - 35.0	S	S	M	M/L	L	M/L	X-L	X-L/XX-L
35.1 - 39.0	M	M/L	L	M/L	X-L	X-L/XX-L	XX-L	X-L/XX-L
39.1 - 43.0	L	M/L	X-L	X-L/XX-L	XX-L	X-L/XX-L	XX-L	X-L/XX-L

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(Footwear covers)

Combat boot size	SCALP Footwear covers
Up to 8	S
8 to 11	M
above 11	L

3.4 Design. The suit consists of a poncho with attached hood, pull on trousers, and pull on footwear covers. The hood and trouser waist are equipped with draw cords and cord-locks. Wrists, ankles, and footwear tops have latex closures.

3.5 Patterns. Standard patterns, which provide a 1/4 inch seam allowance for joining all seams will be furnished by the Government. The Government patterns, which show directional lines and markings for proper assembly, shall not be altered in any way and are to be used as a guide for cutting contractor's working patterns. The working patterns shall be identical to the Government patterns except that additional notching to facilitate manufacture is permissible.

3.5.1 Pattern parts. The pattern parts and applicable material for cutting the suit components shall be as specified in table II.

TABLE II. List of pattern parts

Material	Nomenclature	Cut parts
Coated cloth	Hood	2
	Body	4
	Trouser	2
	Footwear	4
Sole material	Sole	2

3.6 Construction. The construction shall conform in all respects to the requirements specified in table III and herein.

3.6.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table III. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur.

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3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Any thread break or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area and continue over the defective area to a point 1 inch minimum beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 504 and 505 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.6.1.1.a and b.

3.6.2 Heat sealing. The strapping for heat sealing all seams and stitching shall be folded so that the running edges will be enclosed and shall have the adhesion tape specified in 3.3.3.2 applied to the inside by means of a hot air machine, 278°C (as measured at bottom) at 2.3 feet per minute or 327 °C (as measured at top) at 2.6 feet per minute. The strapping shall then be heat seal applied to all seams and stitching using the same adhesive tape and same means of application. The strapping shall be one continuous length for each seam or stitching. Wherever strapping intersects, they shall be overlapped. Butting or piecing shall not be permitted.

3.6.3 Setting of eyelets. A piece of coated cloth shall be inserted between end item layers for reinforcement before setting the eyelets. The eyelets shall be securely clinched without splitting in a manner that will prevent detachment from or cutting of the surrounding material.

3.7 Manufacturing operations requirements. The suit shall be manufactured in accordance with all operations specified in table III. The contractor is not required to follow the exact sequence of operations. The use of drill holes is prohibited.

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3.7.1 Repairs. Repairing the suit is not allowed and at no time is the removal of heat sealing tape permitted. No more than one layer of seam sealing tape shall be permitted in any one area except where seams cross.

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NO.	TABLE III. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. The suit shall be cut in strict accordance with the patterns furnished which show size and notches for proper assembly of all parts and placement of eyelets. All component parts of the suit shall be cut and assembled with the coated side to the outside of the suit. The coated material for the heat seal strapping shall be cut 1-1/8 to 1-1/4 inch wide strips, then folded as directed to 1/2 to 5/8 inch wide.						
2.	Shade and size marking. Mark, ticket, or bundle all components to ensure uniform shade and size throughout the suit.						
3.	Replacement of defective or damaged parts. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.						
4.	Assemble footwear. a. With coated sides together and edges aligned, position two footwear panels together (for each foot) and stitch them together at the front and back seams catching top edge of label in the back seam.	504 or 505	SSa-1	12-14	50	50	

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1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

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TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Assemble footwear. (cont'd)						
	b. With the coated side to the sole material, position sole to the bottom of the footwear and stitch in place.	504 or 505	SSa-1	12-14	50	50	
	c. Heat seal the front and back seam and then the sole seam.						
	d. Position latex elastic at ankle openings and stitch in place.	504 or 505	SSa-1	12-14	50	50	
5.	Assemble trousers.						
	a. With coated sides together and edges aligned, position the trouser fronts to the backs and stitch side seams and crotch seam.	504 or 505	SSa-1	12-14	50	50	
	b. Heat seal side seams and crotch seam.						
	c. Attach eyelets to the front panel per marks on patterns, thread the drawcord thru the eyelets and attach barrel lock. Knot ends together and hot knife the ends.						
	d. Turn the waist to the inside 1 to 1 1/4 inch, inclosing the waist drawcord, and stitch 1/4 to 3/8 inch from edge catching top edge of label at center back.	301	EFa-1	8-10	50	50	
	e. Heat seal the waist hem stitching.						
	f. Position latex elastic at ankle openings and stitch in place.	504 or 505	SSa-1	12-14	50	50	

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TABLE III

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Assemble poncho.						
	a. With coated sides together, position right and left hood panels together and seam together at the back seam.	504 or 505	SSa-1	12-14	50	50	
	b. Heat seal the seam.						
	c. Attach the eyelets to the front neck opening per marks on patterns, thread drawcord thru eyelets, attach barrel locks, and hot knife ends.						
	d. Fold face opening to the inside 1 to 1 1/4 inches, inclosing the drawcord, and stitch 1/4 to 3/8 inch from edge.	301	EFa-1	8-10	50	50	
	e. Heat seal the face opening hem stitching.						
	f. With coated sides together and edges aligned, stitch back seams together, underarm/side seams together, and hood to poncho at the neck seam catching top edge of label in back seam.	504 or 505	SSa-1	12-14	50	50	
	g. Heat seal all seams.						
	h. With coated sides together and edges aligned, stitch front center seam from the bottom of poncho to the bottom of face opening, matching neck seam.	504 or 505	SSa-1	12-14	50	50	
	i. Heat seal center front seam in one continuous length of tape.						
	j. Position latex elastic at wrist openings and stitch in place.	504 or 505	SSa-1	12-14	50	50	

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3.8 Finished measurements. The finished suit shall conform to measurements for each respective size, as shown in table IV.

TABLE IV. Finished measurements

PONCHO	Sleeve length 1/	Back 2/	
Small	29-1/2	31-3/4	
Medium/Large	31	32-3/4	
X-Large/XX-Large	32-1/2	33-3/4	
Trousers	**Out seam 3/**	**Half-waist 4/**	
Small	43-3/4	24	
Medium/Large	46	28	
X-Large/XX-Large	48-1/2	32	
Footwear	**Top relaxed 5/**	**Height 6/**	**Toe height 7/**
Small	6	11	4
Medium	7	12-1/2	4-1/2
Large	8	14	5
	Sole length 8/	**Sole width 9/**	
Small	13	6-1/2	
Medium	14-1/2	7	
Large	16	7-1/2	

Tolerance for all measurements shall be ± 1/4 inch.

1/ Measure from center back to sleeve wrist.

2/ Measure from base of hood to bottom edge of poncho.

3/ Measure from bottom edge of trouser leg to fold line at waist.

4/ Measure from folded edge to folded edge 2 inches below waist.

5/ Measure ankle closure (relaxed) from folded edge to folded edge.

6/ Measure from ankle closure to bottom of sole joining seam.

7/ Measure from top seam to the bottom of the sole seam 3 inches in from from toe tip.

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8/ Measure from the bottom of joining seams for the length of the sole.

9/ Measure from the bottom of joining seams for the width of the sole.

3.9 Performance of adhesion of strapping. After application of the strapping as specified in 3.6.2, the adhesion of strapping seam strength shall be 9.5 (min) lbs./in of width and the hydrostatic pressure (low) shall be 50 cm pressure for 5 min. when tested as specified in 4.4.5.

3.10 Workmanship. The end item shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in table VI.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be accepted as evidence that the components listed below conform to the referenced requirements when tested by the specified method.

Component	Requirement paragraph	Test method
Base cloth	3.3.1	(see table I)
Coated cloth	3.3.3	(see table I)
Latex	3.3.7	---
Cord locks	3.3.8	---

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table V. The lot size shall be expressed in units of suits. The sample unit shall be one suit. The inspection level shall be II (see 6.5).

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TABLE V. End item visual defects

Examine	Defect	Classification Major	Minor
Material	Any cut, hole, or tear	101	
	Any uncoated area	102	
	Any pit, blister, tunnel, or delamination of coating	103	
	Crease or wrinkle resulting in delamination of coating	104	
	Any spot, stain, or streak, more than 1 inch in any direction	105	
	Color not as specified		201
	Any objectionable odor		202
Construction and workmanship	Any component omitted or not as specified	106	
	Any operation omitted or not properly performed	107	
	Any heat sealed area not fully sealed	108	
Seams and stitching	Not as specified	109	
	Stitches per inch less than required		203
	Gage of stitching not as specified		204
	Any raw edge		205
Strapped seams	Any seam or stitching not strapped	110	
	Strapping not securely affixed	111	
	Any blister or delamination of strapping	112	
	Strapping off center by more than 1/8 inch but covers seam or stitching		206
	Any strapping not folded to conceal raw edges	113	
	Any strapping not 5/8 to 3/4 inch after folded and cemented	114	

4.4.4 End item dimensional examination. The end items shall be examined for conformance to all dimensions specified in table IV. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of suits. The sample unit shall be one suit. The inspection level shall be S-3 (see 6.5).

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4.4.5 End item testing. The end items shall be tested for the characteristics listed in table VI. The lot size shall be expressed in units of suits. The sample unit shall be one suit. The inspection level shall be S-2 (see 6.5).

TABLE VI. End item tests

Characteristic	Requirement paragraph	Test method	No determinations per sample unit	Results reported numerically to nearest
Adhesion of strapping: Seam strength	3.9	ASTM D 1682	5	0.1 lb
Hydrostatic pressure (low)	3.9	5516 1/	2/	Pass/Fail

1/ Refers to FED-STD-191.

2/ Once each at the following six locations:
-hood seam
-junction of hood and neck seam
-straight torso seam
-crotch seam
-junction of sole and footwear seam
-leg seam
Evidence of leakage in one or more seam locations shall be considered a test failure.

4.4.6 Packaging examination.

4.4.6.1 In-process test of heat seal closure seam. Test shall be made daily prior to start of filling and sealing the unit packs to determine that heat sealing equipment is producing seals in accordance with the requirements of 5.1.1. One full width heat seal closure seam shall be made on an empty barrier bag using recommended heat seal settings and tested as specified in 4.5.2. Any test failure shall be cause for readjustment of the sealer settings and retested until a satisfactory seam is produced.

4.4.6.2 Unit pack examination. Every filled and sealed unit pack shall be examined for the defects listed below. The finding of any defect shall be cause for rejection of the unit pack. Ponchos and trousers or footwear covers from rejected unit packs may be re-unit packed as specified in 5.1.1 and submitted for reexamination.

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Defect
Volume exceeds 4000 cm3.
Closure seal width not as specified.
Closure seal not continuous.
Tear, cut, or hole in unit pack.
End item damaged by heat seal.

4.4.6.3 Unit pack leakage testing. Every filled and sealed unit pack, prior to packing as specified in 5.2, shall be tested as specified in 4.5.1 for conformance to the leakage requirements in 5.1.1. Any unit pack which fails the test in 4.5.1 shall be opened and the poncho and trousers or footwear covers shall be removed, thoroughly dried, re-unit packed as specified in 5.1.1, and retested as above.

4.4.6.4 Shipping container examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.5).

Examine	Defect
Marking (exterior	Omitted; incorrect; illegible; of improper size; location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container Inside dimensions of container not as specified
Content	Number of unit packs per container less than required Size shown on one or more unit packs not as specified on container 1/

1/ For this defect, one unit pack from each container in the sample shall be examined.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 (see 6.5).

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Loads not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Leakage test. Each unit pack shall be tested for leaks in accordance with the hot water technique of Method 5009 of FED-STD-101. All unit packs shall be conditioned at ambient conditions for at least 4 hours prior to performing the hot water test. The time that the item and all processing materials have been maintained at ambient conditions prior to or during the processing period may be considered a part of the conditioning time. Bubbles which may appear on the surface of the unit pack, but are not released at a slowly decreasing rate, are not to be construed as an indication of failure. A steady stream or recurring succession of bubbles from any surface or seam shall be considered a failure.

4.5.2 Sealed seam test. Test shall be performed in accordance with Method 2024 of FED-STD-101 at normal room temperature with a static load of 50 ounces using fabricated bag specimen procedure. A 5 percent reduction in static load is permitted when room temperature in the test area exceeds 90 degrees F.

4.5.3 Volume test. The volume of the packaged suit shall be determined by submerging it under water and calculating the water displacement in cubic centimeters.

5. PACKAGING

5.1 Preservation Preservation shall be level A.

5.1.1 Level A. One poncho and one pair of trousers of the same size or one pair of footwear covers, shall be unit packed in a 0.004 inch polyethylene bag conforming to type II, style I, of PPP-B-26. The poncho and trousers shall be folded so that cord-locks are not exposed. The outer surfaces of the bag shall be uniformly colored overall with a color in the range of 34079 to 34087, inclusive, of FED-STD-595. The inside dimensions of the bag shall be 11-1/2 inches by 8 inches for the poncho and trousers, and 8 inches by 5-1/2 inches for the footwear covers.

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The bag shall be closed by expelling excess air by physical compression and heat sealing with a 1/2 ±1/8 inch width seal for jaw type sealers, and 3/16 ±1/16 inch width seal for dielectric, impulse or ultrasonic type sealers. The seal shall not damage the poncho and trousers or footwear covers. The final seal shall be straight, continuous, parallel, and close to the edge of the bag. The filled and closed unit shall not leak when tested in accordance with 4.5.1. Closure seals shall show no separation when tested in accordance with 4.5.2. The total volume of the packaged poncho and trousers and footwear covers shall not exceed 4000 cubic cm, when tested in accordance with 4.5.3.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2).

5.2.1 Level A packing. Ten ponchos and trousers, or ten pairs of footwear covers preserved as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of the shipping container shall be approximately 15 1/2 inches in length, 11 1/2 inches in width, and 14 inches in depth for the poncho and trousers, and 11 inches in length, 8 inches in width and 7 1/2 inches in depth for the footwear covers. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, waterproofed, and reinforced with strapping or tape banding in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-STD-147E for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Ten ponchos and trousers, or ten pairs of footwear covers preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of the shipping container shall be approximately 15-1/2 inches in length, 11-1/2 inches in width and 14 inches in depth for the poncho and trousers, and 11 inches in length, 8 inches in width and 7-1/2 inches in depth for the footwear covers. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method
II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

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5.3 Palletization. When specified (see 6.2), ponchos and trousers, or footwear covers, packed as specified in 5.2.2, shall be palletized on a 4 way entry pallet in accordance with load type 1a of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I,II,III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 26 for the poncho and trousers and number 25 for the footwear covers in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contractor or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129. Additionally, unit packs shall be marked as specified in 5.4.1. All unit pack markings, except for fit and size prediction legend and caution markings, shall be marked on the same face of the bag. Fit and size legend shall be marked on the reverse face.

5.4.1 Fit and size prediction legend markings. Each unit pack shall be marked with a fit and size prediction legend as specified in 3.3.13, an instruction legend as specified in 3.3.12 and the following CAUTION notes:

For the poncho/trousers package: CAUTION should be exercised when worn in hot environments to help prevent heat stress

For the footwear cover package: CAUTION: Footwear cover soles may be slippery on wet metallic surfaces

Markings shall be printed in black, bold face characters. The item identification characters shall be a minimum of 3/16 inch in height and the instructive information characters shall be a minimum of 3/32 inch in height.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The suit is intended for use by members of the Department of Defense for liquid contamination avoidance.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.3, and 6.3).

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e. Levels of packing (see 5.2).
f. Type and class of unit load (see 5.2.1).
g. When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL STD 105.

6.5.1 For end item visual examination. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 4.0 for major defects and 15.0 for total (major and minor combined) defects is recommended

6.5.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.5.3 For end item testing. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.5.4 For packaging examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

6.5.5 For palletization examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.6 Source. A suggested source of supply for the cord locks is:

ITW
230 Gerry Drive
Wood Dale, IL 60191

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6.7 Subject term (key word) listing.

Footwear cover
Outerwear
Poncho
Protective trousers

Custodians:
Army - GL
Air Force - 99

Review activities:
Army - EA, MD
Air Force - 82
DLA - CT

Preparing activity:
Army GL

(Project 8415-0619)

*U.S. GOVERNMENT PRINTING OFFICE: 1990--704-034/30079

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-44384	Suit, Contamination Avoidance, Liquid Protection (SCALP)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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